

Thymulin & Thymosin alpha-1 for Healthspan Available for Research Use Only

(Thymulin 0.25 mg daily plus Tα1 0.8 mg three times weekly)

Zn-thymulin modulates T-cell maturation and likely reduces proinflammatory signaling in appropriate inflammatory contexts. Thymosin alpha-1 (Tα1) is a thymic peptide that modulates innate and adaptive immunity. Both plausibly counter immunosenescence and extend healthspan.

The biological rationale is meaningful but unevenly supported. Thymulin has a close relationship to thymic endocrine aging and emerging experimental evidence in age-associated myeloid inflammation. Tα1 has been shown to increase monocyte HLA-DR, CD3+/CD4+ T-cell measures and antigen presentation, potentially restoring the immune system's ability to recognize and attack pathogens, plausibly supporting healthspan.

💧💧 Individually both compounds are probably safe, but safety of the combined Thymulin and thymosin alpha-1 longevity regimen is unknown.

★ Plausibly effective for healthspan extension.

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

This document was updated 11-Sep-26. The most recent version of this document is available at <https://pdfs.researcher6076.com/pdfs/thymulin-plus-thymosin-alpha-1-for-longevity.pdf>



Aging, Longevity and the Immune system

The aging immune system is characterized by reduced production of naïve T cells, changes in T-cell repertoire and differentiation, altered innate-cell function, and impaired responses to some infections and vaccines. Thymic involution contributes by reducing the environment available for new T-cell development. Immune aging also involves peripheral tissues, chronic antigen exposure, metabolism, and the inflammatory environment. [1]

Inflammaging is the persistent, low-grade inflammation associated with aging. Senescent cells, tissue injury, changes in the microbiome, chronic infections, and altered innate signaling all contribute. In aging, reduced protective immunity and excessive inflammation often coexist: an older immune system responds poorly to a new pathogen while maintaining inappropriate background inflammatory activity. The objective of this protocol

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is to improve regulation and useful responses, not to generally increase immune activation. [1,2]

Thymic endocrine decline supplies an additional rationale. A thymus-derived circulating signal can influence cells outside the gland. Administering such a signal could potentially affect peripheral immune regulation despite substantial thymic involution.

Better vaccine responses, fewer clinically important infections, improved recovery from illness, and reduced inflammatory tissue injury are plausible intermediate objectives. Sustained preservation of function and freedom from disease would be healthspan outcomes.

Thymulin Biology and Evidence for Supporting Aging

Thymulin is a thymic epithelial-cell-derived peptide. The zinc-free peptide is often called facteur thymique sérique (FTS); its zinc-bound form, Zn-thymulin, has the characteristic biological activity described in classical thymulin assays. [3] *(In this survey, “thymulin” is the general name used in the literature. “Zinc-free FTS” and “Zn-thymulin” distinguish between the two when that difference matters.)*

Thymulin Human Observations and Zinc interventions

In a crossover study of 44 institutionalized older adults, 20 mg/day of supplemental zinc partially restored serum thymulin activity. The study supports a contribution from zinc status to diminished activity in that population. [5] The replacement-like hypothesis is that aging can reduce an endocrine signal involved in immune regulation and that exogenous active thymulin might compensate for part of that loss.

Thymulin Aging-specific Experimental Findings

Directly on point, Kanemaru 2026 examined age-associated inflammatory myeloid-cell activity using human immune-cell experiments and interventions in aged mice. Thymulin reduced inflammatory myeloid-cell phenotypes in experimental systems and implicated canonical NF-κB signaling. The in vivo tumor experiments used thymulin acetate with equimolar zinc, administered intraperitoneally at 1.5 mg/kg daily for four weeks. In aged mice they found improved tumor control and responsiveness to anti-PD-L1 treatment. [6]

The human-cell component strengthens relevance to human aging biology but remains an ex vivo intervention. The animal component provides causal intervention evidence in the models tested. [6]

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Thymulin Peripheral and Intrathymic Actions

In older experiments Zn-thymulin restored impaired NK activity in cells from old mice in vitro; zinc-free FTS and zinc alone did not reproduce that result. [7] In hydrocortisone-treated aged mice, 100 ng/day of Zn-thymulin for five days modestly improved subsequent splenocyte responses. [8]

These findings suggest administered thymulin has effects beyond T-cell maturation inside the thymus. Peripheral T-cell, NK-cell, and myeloid regulation are relevant to the plausible support Thymulin might offer for aging.

Tα1 Biology and Evidence for Supporting Aging

Tα1 has effects on innate and adaptive immunity. Experimental work identifies dendritic-cell and Toll-like-receptor-associated signaling, including MyD88-dependent pathways. Other work shows induction of dendritic-cell tryptophan catabolism through indoleamine 2,3-dioxygenase, contributing to a regulatory environment. These observations explain why it is simplistic to describe Tα1 as just a generic immune stimulant. [13,14]

Tα1's relevant proposed actions, many occurring in peripheral cells, include antigen presentation, T-cell responsiveness, NK-cell function, and regulation of inflammatory responses. Tα1 may also influence measures associated with thymic output in particular disease settings.

Human evidence relevant to older adults

A double-blind trial in 90 men aged 65–99 years evaluated Tα1 with influenza vaccination, using 900 micrograms/m² subcutaneously twice weekly for eight doses. This is direct human intervention evidence in older adults, but its scope is vaccine antibody response. [15]

Human hepatitis, cancer, and critical-illness studies provide additional information about exposure and disease-specific outcomes. They have different baseline immune abnormalities, concomitant treatments, and therapeutic objectives.

The human evidence for Tα1 is therefore broader than for thymulin, but its aging rationale is plausible.

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Thymulin Dose and Frequency Evidence

Human treatment experience

Human thymulin treatment research is limited. Two randomized, double-blind rheumatoid-arthritis trials compared thymulin schedules of 1, 5, and 10 mg/day; the report favored 5 mg on its clinical assessments and described minimal adverse effects. [9]

Animal dose range and repeated exposure

Animal studies report activity across markedly different doses and disease states. The table below shows the individual experimental contexts.

Study and evidence	Preparation and route	Dose and exposure pattern	Measured finding
Okamoto 1993 [10]; stressed mice	Thymulin; route not specified in abstract	1 ng/kg over a 3-day course	Restoration of stress-impaired antibody response; little effect in normal mice
Henriques-Coelho 2008 [11]; pulmonary-hypertension rats	Thymulin with zinc-containing vehicle control; subcutaneous	100 ng/kg daily for 3 weeks	Changes in pulmonary hypertension, inflammatory signaling, and pathology
Saha 1995 [8]; aged hydrocortisone-treated mice	Zn-thymulin; intraperitoneal	100 ng per mouse daily for 5 days (~4–5 mcg/kg)	Modest improvement in splenocyte responses; no improvement in thymocyte responses
Novoselova 2018 [12]; inflammatory mice	Zn-thymulin, free or nanoparticle-bound; intraperitoneal	1.5 mg/kg on days 1 and 5 during an 8-day experiment	Attenuation of inflammatory and stress-related outcomes
Kanemaru 2026 [6]; aged tumor-bearing mice	Thymulin plus equimolar zinc; intraperitoneal	1.5 mg/kg daily for 4 weeks	Improved age-associated inflammatory and antitumor outcomes

The 1-ng/kg and 1.5-mg/kg mouse doses differ by 1.5 million-fold. Daily administration has clearer precedent in several relevant animal experiments than a single large weekly injection. The evidence is not a head-to-head demonstration that daily treatment is superior, of how long each injection acted, or whether the response changed during treatment. [6,8,11,12]

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Interpretation of the proposed 0.25 mg daily dose

The proposed dose is a speculative selection informed by activity at widely differing experimental doses and the precedent for frequent administration. Human thymulin PK is insufficiently characterized to optimize the interval or predict exposure from 0.25 mg. Frequent small pulses are a mechanistic preference.

Ta1 Pharmacokinetics

Pharmacokinetics and the limits of the pulse model

In a nine-volunteer crossover study, Ta1 given subcutaneously at 900 micrograms/m² reached peak concentrations in approximately one to two hours. The elimination half-life was less than three hours, and single-dose and five-day multiple-dose exposures showed no apparent accumulation. Formulation affected exposure. These data support transient circulating exposure, rather than sustained concentrations throughout a several-day dosing interval. [16]

A short plasma residence time is not the same as cessation of biological activity. Gene expression, cell differentiation, and immune responses can outlast the peptide's measurable presence. Short PK does not establish the optimal interval or prove that three injections are better than two.

Concentration response and clinical dose comparisons

An experiment in normal human lymphocytes found a bimodal concentration-response pattern, with activity around 10^{-12} M and 10^{-8} M and variability between donors, but this is not a universal human therapeutic curve or informative of a subcutaneous dose. [17]

The Arase pilot randomized 16 HBV patients to 0.8 or 1.6 mg using the same schedule; the difference in response was not statistically significant. The larger lino trial enrolled 316 HBV patients and reported broadly similar overall efficacy between doses, with an advantage for 1.6 mg in an advanced-fibrosis subgroup. [18,19]

In metastatic melanoma, a randomized study compared Ta1 doses of 1.6, 3.2, and 6.4 mg with chemotherapy-based combinations. Outcomes did not establish a simple proportional improvement with increasing dose. [21]

High-dose studies include 8- and 16-mg exposures. Such observations concern tolerability and exposure under specific conditions. [28]

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Study and status	Ta1 dose and schedule	Duration	Relevance and limitation
Arase 2003 [18]; randomized HBV pilot	0.8 or 1.6 mg, 6 times weekly for 2 weeks, then twice weekly for 22 weeks	24 weeks of treatment	Direct 0.8-mg exposure; frequent dosing was limited to induction
Iino 2005 [19]; randomized HBV trial	0.8 or 1.6 mg, same induction and maintenance pattern	24 weeks of treatment	Larger human dose comparison; not an aging trial
Rodríguez-Torres 2008 [20]; published secondary HCV analysis	Methods list 0.8, 1.6, or 3.2 mg or placebo 3 times weekly with peginterferon	Planned course of 48 weeks	Specific 0.8-mg schedule precedent; thyroid-focused report does not establish dose-specific efficacy or completion for every participant
Gravenstein 1989 [15]; randomized vaccine trial	900 micrograms/m ² subcutaneously twice weekly	8 doses	Older-adult intervention; different dose basis and outcome
ETASS 2013 [23]; randomized sepsis trial	1.6 mg every 12 hours for 5 days, then daily for 2 days	7 days	Acute intensive exposure; not long-term aging evidence
TESTS 2025 [24]; phase 3 sepsis trial	1.6 mg every 12 hours	Up to 7 days	No clear reduction in primary mortality outcome; short-term safety context
Qiu 2015 [27]; published HCC trial protocol	1.6 mg twice weekly after resection	Planned 12 months	Describes intended treatment; a protocol alone supplies no completed-outcome evidence

In early advanced-cancer research, immune effects often emerged after 24–48 hours and seldom persisted beyond 72 hours, motivating a proposal for repeated treatment every two to three days. [22]

The proposed 0.8 mg three times weekly provides 2.4 mg/week. A 1.6-mg twice-weekly regimen provides 3.2 mg/week. Thus, the proposal combines smaller individual doses and more frequent administration with a lower weekly total.

Mechanistic Rationale

The combination hypothesis is that thymulin could address diminished thymic endocrine signaling and maladaptive peripheral inflammatory activity, while Ta1 could support antigen presentation and coordination of innate and adaptive responses. Both may affect inflammation and T-cell or NK-cell function.

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11-Sep-26*

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The combination does not require a sequence of “maturation first, activation second.” T-cell development is not shown to accelerate on a one-day timetable after the proposed thymulin dose, and both peptides can influence peripheral cells.

The proposed benefits are improved responses to new antigens or vaccines, reduced clinically important infection burden, better immune recovery after stress, and less damaging chronic inflammatory activity. Improved surveillance of abnormal cells is another mechanistic possibility.

Proposed Longevity and Healthspan Dose

The following is proposed for healthspan and to support aging.

Component	Proposed amount and frequency	Weekly total	Basis and principal uncertainty
Thymulin	0.25 mg daily	1.75 mg	Frequent-dose experimental precedent; the human dose, active formulation, exposure, and aging benefit are unestablished
Ta1	0.8 mg three times weekly	2.4 mg	Human 0.8-mg exposure and schedule precedent; efficacy in aging and benefit of combining it with thymulin remain unestablished

No loading phase, dose escalation, fixed one-day sequencing rule, or drug holiday is established by the evidence for this combination. Animal studies and 24- or 48-week disease-treatment schedules do not select an appropriate human healthspan course.

Plausible Longevity / Healthspan Benefits

- **Plausibly improved adaptive responses to new pathogens and vaccines:** Ta1 likely supports antigen presentation and T-cell activation, while Zn-thymulin’s role in T-cell maturation could complement these effects.
- **Plausibly reduced susceptibility to or severity of some infections:** Complementary effects on T-cell maturation, antigen presentation, and immune-cell responses could support pathogen clearance and limit infection-related inflammatory damage.
- **Plausibly improved immune surveillance of abnormal cells:** Complementary effects on T-cell maturation, antigen presentation, and immune-cell function could support recognition and elimination of abnormal cells.
- **Plausibly reduced age-related chronic inflammation and associated tissue damage:** Zn-thymulin and Ta1 could complement one another in regulating

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inflammatory signaling and immune responses, potentially limiting persistent inflammation and the tissue damage it causes.

- **Plausible support for immune recovery after acute physiological stress:**
Complementary effects on T-cell maturation and immune regulation could help restore immune function when acute illness or injury has disrupted it.

Cautions

- **Active autoimmune disease with unstable control**, theoretical risk of immune modulation worsening autoimmunity. (Tα1 has sometimes been explored in autoimmune contexts with mixed results.)
- **Zinc status and supplementation:** Adequate zinc availability supports Zn-thymulin activity.
- **Pregnancy and breastfeeding.**
- **Cancer-treatment coordination.**
- **Immunosuppressants and biologics.**

Drivers of Diminished Response and Mitigation Strategies

Tα1 has clinical use over weeks and months without observing diminished responsiveness, but that is no proof that longer or more frequent use may not lead to waning. [16,18–24,27]

For thymulin, biological effects were observed after repeated administration in animal studies over days to weeks. The older human arthritis studies add treatment experience, but do not prove adaptation during chronic low-dose aging does not occur. Demonstrating an effect at an experiment's endpoint is weaker evidence than measuring responsiveness to successive doses throughout treatment. [6,8–12]

Available evidence does not show clinically important progressive desensitization for either proposed component, but it is insufficient to exclude it.

Zinc Supplementation

Zinc is an essential cofactor for Thymulin activity and a central micronutrient for innate and adaptive immunity. Adequate zinc status supports **thymic hormone function, thymocyte maturation, T-cell development, and multiple zinc-dependent enzymes and transcription factors** that regulate immune signaling. **Adequate zinc availability could also support formation of active Zn-thymulin after administration of zinc-free Thymulin.**

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Population data show **zinc deficiency or insufficiency is relatively common in older adults** because of reduced intake, absorption changes, polypharmacy, and chronic disease; this makes zinc assessment and correction a logical adjunct when targeting thymic regeneration or immune support. **Dietary reference / typical targets:** the U.S. Recommended Dietary Allowance (RDA) is **about 11 mg/day for adult men and 8 mg/day for adult women. The adult tolerable upper intake level is 40 mg of elemental zinc daily from food, supplements, and medications combined.** Chronic excessive intake can cause copper deficiency, anemia, low neutrophil counts, and neurological or immune problems.

Serum zinc, serum copper, and ceruloplasmin may help assess zinc-copper balance and inform supplementation decisions when interpreted alongside dietary intake and clinical context.

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